Brian Kyanjo

Computational Scientist | Research Software Engineer

HPC & Scientific Computing | Numerical Modeling | Open-Source Research Software

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Professional Summary

Computational scientist and scientific software developer specializing in high-performance scientific computing, numerical modeling, data assimilation, and scalable research software. My work focuses on developing reliable, reproducible, and interoperable computational systems for large-scale scientific applications. I have experience developing open-source scientific software, MPI- and GPU-based workflows, HPC and cloud computing platforms, containerized environments, and coupled modeling systems across Python, MATLAB, and compiled scientific codes.

Education

Boise State University (BSU) *2024*

Computing Ph.D. in Computational Math Sciences and Engineering

Dissertation: *GeoFlood: A Computational Model for Overland Flooding*

Advisor: Prof. Donna Calhoun

African Institute for Mathematical Sciences (AIMS), Rwanda *2020*

Master of Science in Mathematical Sciences (Climate Resilience)

Thesis: *Simulating Tropical Cyclone over the Southwest Indian Ocean using Model for Prediction Across Scales (MPAS): Sensitivity to Cloud Microphysics Schemes*

Advisor: Prof. Babatunde J. Aboidun

Makerere University (MAK) *2018*

Bachelor of Science in Physical Sciences (Physics and Mathematics)

Final Project: *Water Level Sensor System (Design and Construction)*

Advisor: Prof. Ireeta Tumps Winston

Technical Skills

Scientific Computing: Numerical PDEs, finite volume methods, adaptive mesh refinement (AMR), data assimilation, ensemble methods, uncertainty quantification, numerical modeling

Programming: Python, C++, C, FORTRAN, MATLAB, CUDA, R, Java, Bash

HPC and Parallel Computing: MPI, OpenMP, CUDA, Slurm, distributed-memory computing, GPU acceleration, parallel I/O, scalable scientific workflows

Scientific Software Engineering: Git, Linux, APIs, model coupling, Python–MATLAB interoperability, testing, workflow automation, reproducible computational workflows

Research Infrastructure: Docker, Apptainer, Spack, Jupyter, AWS Batch, Podman, systemd Quadlet, HPC deployment, containerized scientific environments

Additional: LaTeX, SQL, HTML, electronics/auto technician experience

Professional Experience

Georgia Institute of Technology *Sept. 2026 – Present*
Research Scientist II
Research Software Engineer, OSPO@GT (since September 2026; part-time, 30% FTE allocation October–December 2026).
Contribute to open-source AI platform development for Georgia Tech educators under Dr. Fang (Cherry) Liu, including GT-DARE deployment, operational documentation, and access/onboarding workflow development. This work is part of my Georgia Tech appointment alongside EAS research.

Previous appointment: Postdoctoral Research Fellow – Computational Science and Scientific Software, Aug. 2024–Aug. 2026.

- Design and develop ICESEE, an open-source, MPI-parallel ensemble data assimilation framework for state and parameter estimation in ice-sheet and geophysical models.
- Develop scalable model-coupling interfaces and workflows for integrating ICESEE with independent scientific models, including ISSM and Icepack.
- Parallelize and optimize ensemble simulation and data assimilation workflows for distributed-memory HPC systems using MPI and scalable scientific computing methods.
- Design and develop CryoStack, a modular scientific cyberinfrastructure platform connecting cryosphere data, models, data assimilation, and heterogeneous computing environments through CryoLauncher, ICESEE, Frozen Legacies, and LIVIST.
- Develop Remote and AWS cloud execution capabilities for scientific workflows, including containerized execution on AWS Fargate and EC2 and integration with institutional computing resources.
- Develop reproducible scientific software environments using Docker, Apptainer, Spack, and HPC deployment workflows.
- Develop Python–MATLAB interoperability and ISSM–ICESEE coupling workflows for ensemble forecasting, parameter estimation, and data assimilation.
- Develop workflow automation for experiment configuration, provenance, simulation diagnostics, HPC job orchestration, and scientific result management.
- Contribute to research on scalable data assimilation and computational methods for understanding ice-sheet dynamics and their contribution to future sea-level change.

Georgia Institute of Technology *Apr. 2025 – Present*
Research Faculty Advisory Council (RFAC), EAS

- Represent research faculty interests within the College of Sciences.
- Serve as a liaison between research faculty and college leadership.
- Advance initiatives supporting faculty engagement, inclusion, and professional growth.
- Evaluate internal grant proposals and contribute to funding decisions.

Rocky Mountain Advanced Computing Consortium (RMACC) *Oct. 2023 – Sept. 2024*
RMACC Apprenticeship

- Designed, quoted, and deployed a major expansion of the regional high-performance computing cluster at the University of Colorado Boulder (Alpine supercomputer).
- Designed a research computing cluster in a Unix environment.
- Supported HPC cluster provisioning, deployment automation, and systems administration workflows.
- Created playbooks and container images for MPI-based applications.

Boise State University, School of Computing *June 2023 – Dec. 2023*
Graduate Teaching Assistant

- Developed a computational model for overland flooding: github.com/KYANJO/GeoFlood.
- Developed a CUDA-based version of GeoFlood.
- Served as teaching assistant for Introduction to Programming in Mathematics.

Boise State University, School of Computing *Aug. 2020 – July 2024*
Graduate Research Assistant

- Developed GeoFlood, a computational model for overland flooding: github.com/KYANJO/GeoFlood.
- Benchmarked GeoFlood across CPU and GPU computing platforms.
- Developed CUDA-based Riemann solvers for GeoFlood.
- Developed ForestClaw extensions for atmospheric applications.
- Explored data-driven and machine learning approaches to enhance predictive modeling and simulation accuracy.
- Implemented a discontinuous Galerkin solver within ForestClaw as a summer research project.
- Completed a comprehensive examination on Riemann solvers for wet/dry interfaces in finite-volume schemes: github.com/KYANJO/Comp_exam/tree/main/code/comp_exam_artifact.
- Implemented approximate and exact Riemann solvers for handling wet/dry interfaces using finite-volume schemes.
- Investigated accelerated wave-propagation algorithms using GPU and CPU comparisons on HPC platforms.

Idaho National Laboratory (INL) *May 2022 – Sept. 2022*
Intern

- Supported post-irradiation examination capabilities for nuclear fuel using gamma tomography.
- Simulated gamma-emission tomography of nuclear fuel and generated tomographic reconstructions to identify isotopic distributions.
- Performed thermal-fluid analyses of geothermal closed-loop configurations and discussed results with the INL research team.
- Participated in team meetings and contributed to maintenance and improvement of the software suite.

AIMS Rwanda *Sept. 2020 – Dec. 2020*
Tutor

- Tutored master's-level students in mathematics.
- Assisted lecturers with academic quality assurance.
- Supported administrative, library, IT, logistics, and technical activities as needed.

ESMT Berlin *2020*
Certificate in Business Analytics

Marga Business Simulations, Germany *2020*
Certificate in Business Simulations Management

- Developed and marketed products within a competitive business simulation.
- Collaborated with six students from different universities and disciplines.
- Served as team lead.

God Cares High School *Jan. 2020 – July 2020*
Physics and Mathematics Teacher

- Taught physics and mathematics.
- Supervised laboratory activities.

Ubuntu Hill School *Aug. 2019 – July 2020*
Physics and Mathematics Teacher

- Taught physics and mathematics.

Kampala Capital City Authority (K.C.C.A.) *May 2017 – Aug. 2017*
Intern

- Fabricated, repaired, and replaced damaged street lights and electronic control boxes.
- Troubleshot, maintained, and installed electric and solar street lights across the city.
- Supported maintenance of traffic lights and investigated approaches for reducing peak-hour traffic.

Selected Funded Projects

- **Arête Glacier Initiative Sliding Law RFP**, *A Framework for Detection of Changing Subglacial Properties through Transient Data Assimilation*, 2026–2028.
  Award: $186,965. PI: Alexander Robel; Co-PI: Talea Mayo; Co-Investigator: Brian Kyanjo. The project focuses on transient data assimilation for detecting changes in subglacial properties and improving constraints on future sea-level projections.

Selected Scientific Software

- **ICESEE (Ice Sheet State and Parameter Estimator)** – Open-source, MPI-parallel ensemble data assimilation framework for state and parameter estimation in ice-sheet and geophysical models. github.com/ICESEE-project/ICESEE

- **CryoStack** – A modular scientific cyberinfrastructure platform connecting cryosphere data, models, data assimilation, and heterogeneous computing environments.
  Visit CryoStack | github.com/ICESEE-project/CryoStack
- **ICESEE-Containers** – Reproducible container infrastructure for deploying MPI-based scientific software across HPC systems using Docker and Apptainer.
  github.com/ICESEE-project/ICESEE-Containers
- **ICESEE-Spack** – Spack-based software environment for reproducible builds and deployment of ICESEE and coupled scientific modeling stacks.
  github.com/ICESEE-project/ICESEE-Spack
- **GeoFlood** – Hybrid CPU/GPU computational model for overland flooding using finite volume methods and adaptive mesh refinement.
  github.com/KYANJO/GeoFlood
- **ForestClaw Extensions** – Numerical solver and adaptive mesh refinement development for PDE-based geophysical simulations.
  github.com/KYANJO/forestclaw
- **GT-DARE – Georgia Tech OSPO contributions** – Deployment and research software engineering through Georgia Tech OSPO: Podman/systemd services, operations documentation, and Django/React access-request and administrator-approval workflows.
  github.com/gt-ospo/gt-dare-deploy

Publications and Manuscripts

- Brian Kyanjo, Alexander A. Robel, Angelo S. Tarzona, Eliza J. Dawson, and Winnie Chu. *CryoStack-v1.0.0: A modular cyberinfrastructure stack for cryosphere data, models, data assimilation, and heterogeneous computing.* Submitted to Journal of Open Source Software (JOSS). Read manuscript.
- Brian Kyanjo and Alexander Robel. *The Ice Sheet State and Parameter Estimator (ICESEE) Library: Ensemble Kalman Filtering for Ice Sheet Models.* Submitted. 10.48550/arXiv.2603.26947.
- Brian Kyanjo, Donna Calhoun, and David L. George. *A Hybrid CPU/GPU GeoFlood: High-Performance Computing in Overland Flooding.* In preparation.
- Brian Kyanjo, Donna Calhoun, and David L. George. *GeoFlood: Computational Model for Overland Flooding.* Submitted. 10.48550/arXiv.2403.15435.
- B. Kyanjo, D. Calhoun, C. Burstedde, S. Aiton, J. Snively, and M. Shih. *GPU Accelerated Adaptive Wave Propagation Algorithm.* 2023. 10.13140/RG.2.2.11304.34569.
- Carlo Parisi, Paolo Balestra, Brian Kyanjo, Theron D. Marshall, Travis L. McLing, and Mark D. White. *Closed Loop Geothermal Analysis Modeling and Simulation Using Idaho National Laboratory's RELAP5-3D-FALCON Coupled Codes.* Proceedings of the 48th Workshop on Geothermal Reservoir Engineering, Stanford University, Stanford, California, February 6–8, 2023, SGP-TR-224.

Selected Conferences and Short Courses

- American Geophysical Union (AGU) Annual Meeting, 2025
- Society for Industrial and Applied Mathematics (SIAM), 5th Biennial Meeting of the SIAM Pacific Northwest Section, 2025
- Scalable and Computationally Reproducible Approaches to Arctic Research, 2025

- Adviser and Icepack Workshop and Hackathon, 2025
- The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC), 2024
- United States Research Software Engineers (US-RSE) Conference, *Software-Enabled Discovery and Beyond*, 2023
- Accelerating Computing for Emerging Sciences (ACES) Workshop, 2023
- SIAM Conference on Parallel Processing for Scientific Computing (PP22), 2022
- SIAM Conference on Computational Science and Engineering, attendee and poster judge
- Winter School, *Advanced Numerical Methods for Hyperbolic PDE*, Laboratory of Applied Mathematics, University of Trento, Italy, 2021

Selected Conference Posters

- *ICESEE: An Open-Source, Scalable Data Assimilation Library for Ice Sheet Models*, AGU 2025
- *GeoFlood: A Computational Model for Overland Flooding*, 2023 Graduate Showcase, Boise State University
- *GPU-Accelerated Wave Propagation Algorithm*, 2023 Research Computing Days Conference
- *Geothermal Closed Loop*, 2022 Clean Energy Development, Idaho National Laboratory
- *GPU-Accelerated Wave Propagation Algorithm*, 2022 3rd Biennial Meeting of the SIAM Pacific Northwest Section

Invited Talks and Visits

- ESCO 2026: *Advanced Computational Methods for Earth System Modeling and Analysis*
- *ICESEE: An Open-Source, Scalable Data Assimilation Framework for Ice Sheet Modeling*, 5th Biennial SIAM North Pacific Meeting
- *Math on ICE: ICESEE Coupling with ISSM for EnKF Data Assimilation*
- Seminar for the Geosystems Group: *Unified Computational Approaches to Flood and Ice Sheet Modeling*
- *Data Assimilation with Ice Sheet Modeling*, ART@SC24
- *GeoFlood: Computational Model for Overland Flooding*, 4th Biennial Meeting of the SIAM Pacific Northwest Section
- *GeoFlood: A Computational Model for Overland Flooding*, 2023 Lightning Talk, ACES Workshop
- Research Visit at USGS: *Overland Flow Modeling with David L. George*, June 2023
- Keynote Speaker, GAIN: Promoting and encouraging study abroad for students from low-developing countries

Activities, Associations, and Awards

- 5th Biennial SIAM North Pacific Meeting – Mini-symposium organizer and poster judge, *Data Assimilation for Earth and Atmospheric Systems* (October 2025)
- Georgia Institute of Technology – Research Faculty Advisory Council Representative for the School of Earth & Atmospheric Sciences (April 2025–Present)

- Vanderbilt University – Adviser & Icepack Workshop and Hackathon Grand Prize Winner
- Boise State University – NSBE Public Policy SIG STEM Education, Team Co-Lead (Oct. 2021 – Oct. 2022)
- Boise State University – NSBE Student Chapter, President (April 2023 – April 2024)
- Boise State University – NSBE Student Chapter, Vice President (April 2021 – April 2023)
- Boise State University – SIAM Student Chapter, President (Mar. 2022 – April 2023)
- 2023 Research Computing Days Hackathon – Organizer and Judge
- Idaho school outreach activities promoting STEM education
- Boise State University – SIAM Student Chapter, Vice President (Mar. 2021 – Mar. 2022)
- 2021 SIAM Travel Award (AN21)
- SIAM Travel Award (GS21)
- Graduate Research Assistantship Scholarship (Ph.D.), Aug. 2020 – July 2024
- 2023 Innovation Award Finalist, Promoting STEM Across Idaho
- Research Computing Days Conference – Winning Poster, *GPU Wave Propagation Algorithm*
- Association of State Floodplain Managers (ASFPM)
- United States Research Software Engineers (US-RSE)
- US-RSE 2023 Conference Travel Award
- AIMS Rwanda – AIMS Master's Scholarship (M.Sc.), Aug. 2019 – July 2020
- ESMT Berlin – Team Lead (May 2020 – July 2020)
- Makerere University Physics Society – Vice President (April 2017 – April 2018)
- Government Scholarship on National Merit (B.Sc.), Aug. 2015 – May 2018

References

- Prof. Alexander Robel, Georgia Institute of Technology – alexander.robel@eas.gatech.edu
- Prof. Donna Calhoun, Boise State University – donnacalhoun@boisestate.edu
- Mr. Craig Earley, University of Colorado Boulder – craig.Earley@colorado.edu
- Dr. David L. George, U.S. Geological Survey – dgeorge@usgs.gov